

5. Nanoelectronic Devices

Quantum confinement - Quantum structures - quantum wells, wires and dots - Zener-Bloch oscillations - Resonant tunneling - quantum interference effects - mesoscopic structures - Single electron phenomena - Single electron Transistor. Semiconductor photonic structures - 1D, 2D and 3D photonic crystal. Active and passive optoelectronic devices - photo processes - spintronics - carbon nanotubes: Properties and applications.

INTRODUCTION

A **nanometre** (nm) is one billionth $\left(\frac{1}{10^9}\right)$ of a metre. For comparison, thickness of a single human hair is about 80,000 nm (80 μm), a red blood cell is approximately 7,000 nm (7 μm) wide and a water molecule is almost 0.3 nm across.

Scientists and engineers are nowadays interested in nanoscale which is from 1 nm to 100 nm. At nanoscale, the properties of materials are very different from those at larger scale. Therefore, the nano-world is in between quantum world and macro world.

Nanoscience

It is concerned with the study of phenomena and manipulation of materials at nanometre scale.

Nanotechnology

It is the design, characterization, production and application of structures, devices and systems by controlling shape and size at the nanometre scale.

Nanotechnology means making use of the unique physical properties of atoms, molecules and other materials measuring roughly 1 to 100 nanometre.

The word “nano” comes from nanos, a Greek word meaning dwarf.

Presently, we are using many devices made of nanoelectronic devices. The microelectronics industry was born out of the invention of the bi-polar transistor in 1947 and by the invention of the integrated circuit (IC) in 1958.

Gordon Moore (co-founder of INTEL Corporation) observed that the number of transistors per square inch on IC chip roughly doubled by every 18 to 24 months. This general rule of thumb is now called as “**Moore’s law**”.

By 1960, the minimum feature size of a transistor was approximately 100 μm . At present, manufacturing technology is at transistor size of 22 nm.

Because of the diminishing feature size of transistors and other components, we can say that the electronics industry is already “doing” nanotechnology.

Nanomaterials

Definition

Nanophase materials are newly developed materials with grain size at the nanometre range (10^{-9} m), i.e., in the order of 1 - 100 nm. The particle size in a nano material is 1 - 100 nm. They are simply called **nanomaterials**.

Different forms of Nanomaterials

Nano-structured material

The structures whose characteristic variations in design length is at the nanoscale.

Nano particles

The particles size in the order of 10^{-9} m are called nano particles.

Nano dots

Nanoparticles which consist of homogeneous material, especially those that are almost spherical or cubical in shape.

Nanorods

Nanostructures which are shaped like long sticks or rods with diameter in nanoscale and a length very much longer.

Nanotubes

Nanotubes are nanoscale materials that has a tube like (hollow cylinder) structure.

Nanowires

Nanowires are solid rod like material with diameter of few nanometers or less.

Fullerenes

A form of carbon having a large molecule consisting of an empty cage of 60 or more carbon atoms.

Nanocomposites

Composite structures whose characteristic dimensions are found at nanoscale.

Cluster

A collection of units (atoms or reactive molecules) upto few tens of units.

Colloids

A stable liquid phase containing particles in the 1-1000 nm range.

Nano electronics

Nanoelectronics refers to the use of nanotechnology in electronic components, especially transistors.

It often refers to transistor devices that are so small that inter-atomic interactions and quantum mechanical properties need to be studied extensively.

Besides, being small and allowing more transistors to be packed into a single chip, the uniform and symmetrical structure of nanotubes allows a higher electron mobility, a symmetrical electron/hole characteristic.

Need for Nanotechnology in Electronics

Today microelectronics are used to solve most of the problems.

The two exceptional disadvantages of micro electronics are:

- Physical size
- Increasing cost of fabrication of integrated circuits.

To overcome these disadvantages, nanotechnology is used.

Advantages of Using Nanotechnology in Electronics

- Increasing the density of memory chips.
- Decreasing the weight and thickness of the screens.
- Nanolithography is used for fabrication of chips.
- Reducing the size of transistors used in integrated circuits.
- Improving display screens on electronic devices.
- Reducing power consumption.

5.1 QUANTUM CONFINEMENT

Definition

It is a process of reduction of the size of the solid such that the energy levels inside become discrete.

In this case, small “droplets” of isolated electrons are created. Thus, the energy of a small volume of such materials

are quantized just like in an atom. This type of artificial or fake atoms have **tunable** electrical properties.

The quantum confinement effect is observed when the size of the particle is too small to be comparable to the wavelength (de-Broglie wavelength) of the electron.

It is to be noted that only a small percentage of electrons that are free to move during confinement and majority of electrons still tightly bound within inner orbitals.

Usually in order to reduce the dimensions of a given volume, either bottom-up approach or top-down approach is followed.

In bottom-up approach, low-volume structures are built atom by atom. In the top-down approach, material is removed from one or more of three dimensions (length, width, height) of a larger solid.

In both cases, a structure small enough for quantum behavior to manifest can be produced.

5.2

QUANTUM STRUCTURES (Quantum Wells, Wires and dots)

Definition

When a bulk material is reduced in its size, atleast one of its dimension, in the order of few nanometres, then the structure is known as quantum structure.

Explanation

The volume of a box can be reduced by shortening its length, width or, and height. The same is true for the region occupied by the electrons in a solid.

There are three dimensions to confine the bulk material. The quantum confinement needs confining at least one of these

dimensions to less than 100 nanometres or even just a few nanometres.

The more the dimensions are confined, the more the density of states function looks like that of an atom. This progressive **discretization** gives new ways to understand real atoms, behavior of electrons and developing quantum confined electronic devices.

A structure in which the motion of the electrons or holes are confined in one or more directions by potential barriers is called quantum confined structure.

The quantum confined structure is classified into three types based on the confinement directions. They are

- (i) **quantum well**
- (ii) **quantum wire**
- (iii) **quantum dot**

(i) Quantum well (2 - dimension)

Definition

When the electrons are confined inside a region of minimal width i.e., confinement in one dimension “**quantum-well**” is created.

In other words, if one dimension is reduced to the nanometre range while the other two dimensions remain large, then we get a structure known as **quantum well**.

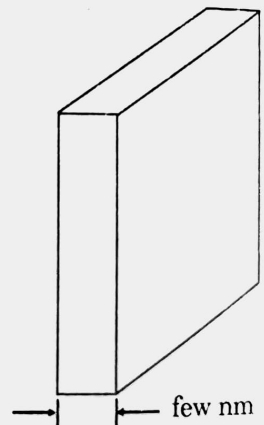


Fig. 5.1 Quantum Well (2D)

Fig. 5.1 shows a quantum well or 2-D structure

Construction

Quantum wells are made from alternative layers of different semiconductors or by deposition of very thin metal films.

Explanation

The quantum well is a larger structure in which the carrier particles are free to move in two dimensions. The particles are confined in one dimension hence, they are considered to be **quantum confinement**.

Due to the confinement of carriers, the quantum well structure has important applications in making useful devices.

Use

Quantum wells are now widely used to make semiconductor lasers and other important devices.

(ii) Quantum wire (1 dimension)

Definition

When the electrons are confined in two mutually perpendicular directions, then the structure is known as quantum wire.

In other words, if two dimensions are reduced and one remains large, the resulting structure is quantum wire.

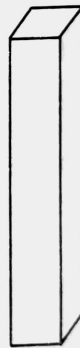


Fig. 5.2 Quantum wire (1D)

Fig. 5.2 shows a quantum wire or 1-D structure

Explanation

The carriers trapped in such structures are considered to be in 1-D quantum confinement. In this case, a carrier is only free to move its trajectory along the wire.

Example

Examples of quantum wire structures are nanowires, nanorod and nanotube.

(iii) Quantum Dots (0 - dimension)

Definition

When all the three dimensions are minimized, the resulting structure is known as quantum dot.

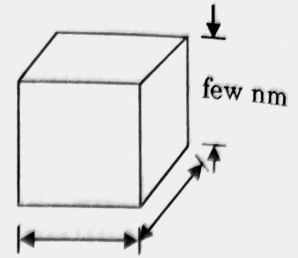


Fig. 5.3 Quantum dot

The dot can be particle located inside a larger structure or on its surface. It can also be a place where electrons have been trapped using electrical fields.

Fig. 5.3 shows a quantum dot or 0-D structure

Explanation

In quantum dot, the carriers has only confined states i.e., there are no freely moving. Though a quantum dot has many thousands of atoms, carriers it is considered more like a single atom due to its peculiar properties.

Use

Quantum dot can be used in a quantum computer and quantum dot lasers etc.

Fig. 5.4 and Fig. 5.5 explains the processes of diminishing the size in the case of rectilinear geometry and curvilinear geometry, respectively.

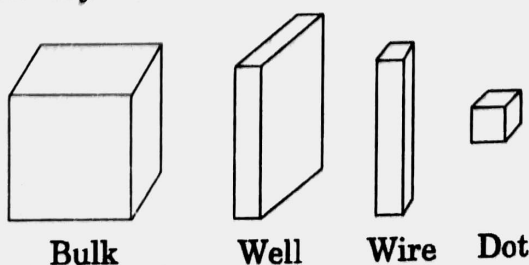


Fig. 5.4 Rectangular nanostructures

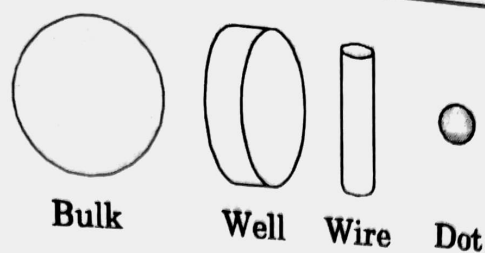


Fig. 5.5 Curvilinear nanostructures

Fig. 5.6 shows a comparison of three quantum confined structures with bulk material.

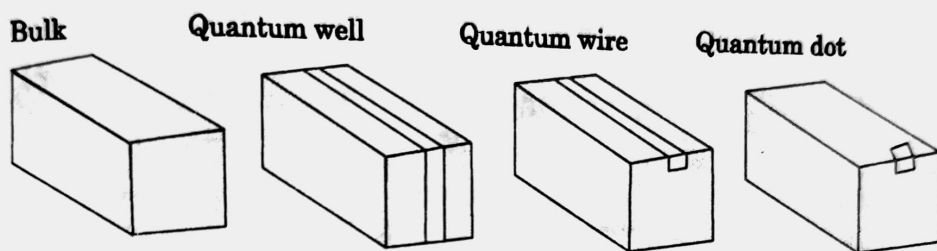


Fig. 5.6 Three quantum structures

As quantum wells and quantum wires each have at least one dimension in which the electrons are free to move and these structures exhibit **“partial confinement”**. However, quantum dots exhibit **“total confinement”**.

The classification of quantum confined structures is shown in **Table 5.1**.

Table 5.1

Structure	Quantum confinement directions	Number of unconfined dimensions
Bulk	0	3
Quantum well	1	2
Quantum wire	2	1
Quantum dot	3	0

Density of states in Quantum Well, Quantum Wire and Quantum Dot Structure

Bulk Structure

The density of state of a bulk material is given by

$$Z(E) = \frac{8 \pi \sqrt{2} m^{*3/2} (E - E_c)^{1/2}}{h^3}$$

where E_c - bottom of conduction band energy

m^* - effective mass of electron

The density of state curve is parabolic as shown in fig. 5.7(a).

Quantum Well Structure

In quantum well structure, the electrons can move freely in two directions (2D) and confined in only one direction. The density of state of the quantum well structure

$$Z(E) = \frac{4 \pi m^*}{h^2} \quad E_0 > E_i, \quad i = 1, 2, 3$$

The density of state function is constant and it gives step like function with energy as shown in fig. 5.7 (b).

Quantum Wire Structure

The quantum wire structure provides only one non confinement direction, i.e., the carrier can move freely along one direction (1D). The remaining two directions are confined for charge carrier.

The density of states of quantum wire is proportional to $\frac{1}{\sqrt{E}}$. The density of state for this structure is given as

$$Z(E) = \frac{2\sqrt{2} m^* (E - E_i)^{-1/2}}{h} \quad i = 1, 2, 3$$

The density of states function with energy is as shown in fig. 5.7(c).

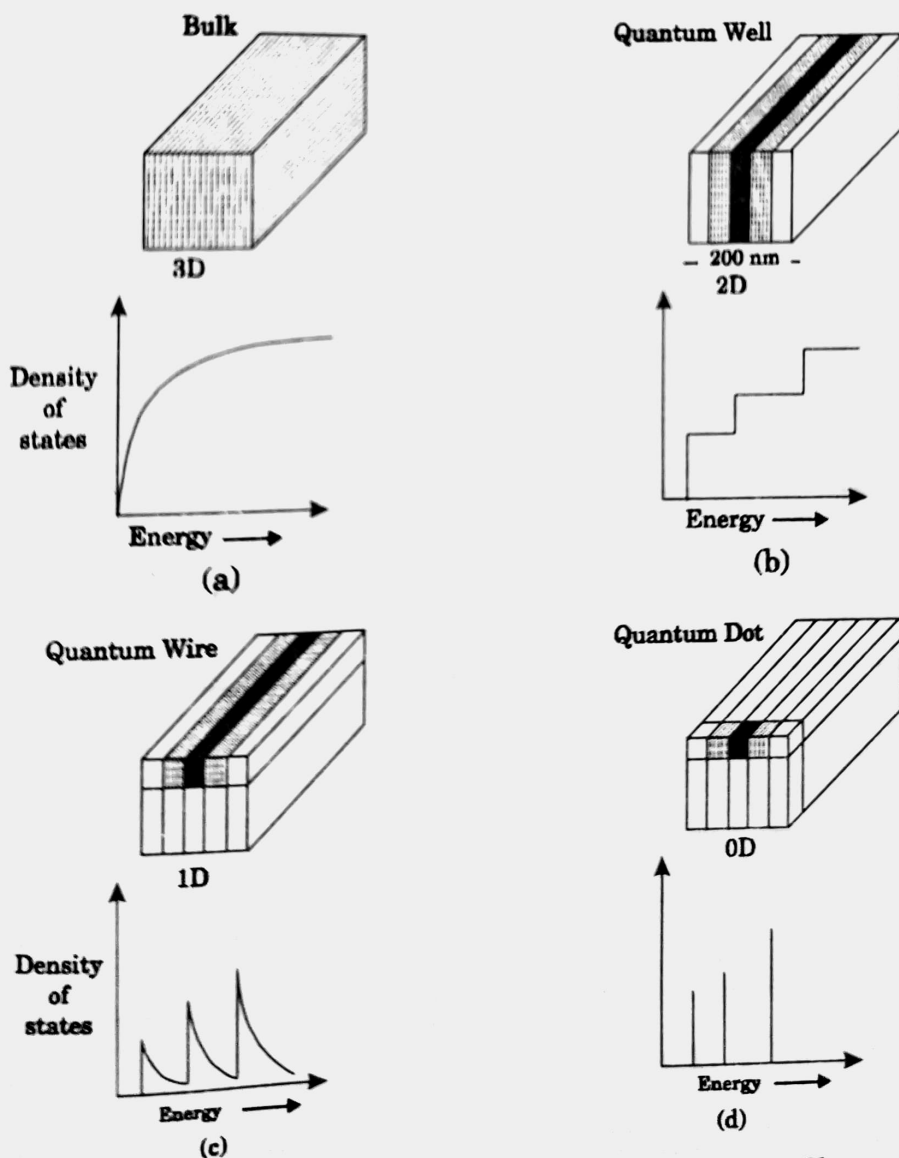


Fig. 5.7 Structure and Density of states for bulk, quantum well, quantum wire and quantum dot

Quantum dot

In a quantum dot all the directions are confined and no direction (0D) in which electron movement is free. The density of state for this structure is given by

$$Z(E) = \delta(E - E_i), \quad i = 1, 2, 3$$

For quantum dot (0D) or nano particle, the permitted energy values are not continuous but form discrete bunches of varying densities. Thus, density of energy levels appear as discrete line as in fig. 5.7(d).

Table 5.2 gives different quantum structures and their corresponding density of state functions.

Table 5.2

Type of structures & No. of non confinement Dimensions	Density of state function
Bulk (3D)	$Z(E) = \frac{8\pi \sqrt{2} m^{*3/2} (E - E_c)^{1/2}}{h^3}$
Quantum well (2D)	$Z(E) = \frac{4\pi m^*}{h^2}; E > E_i, \quad i = 1, 2, 3$
Quantum wire (1D)	$Z(E) = \frac{2\sqrt{2} m^* (E - E_i)^{-1/2}}{h}, \quad i = 1, 2, 3$
Quantum dot (0D)	$Z(E) = \delta(E - E_i), \quad i = 1, 2, 3$

5.4

RESONANT TUNNELING

T

Definition

The phenomenon in which the tunneling current reaches peak (maximum) value, when the energy of incident electron wave is equal to quantized energy state

of the quantum well formed by the double symmetric barriers is known as resonant tunneling.

Explanation

An interesting phenomena occurs when two barriers of width a separated by a potential well of small distance L as shown in fig. 5.8. This leads to the concept of *resonant tunneling*.

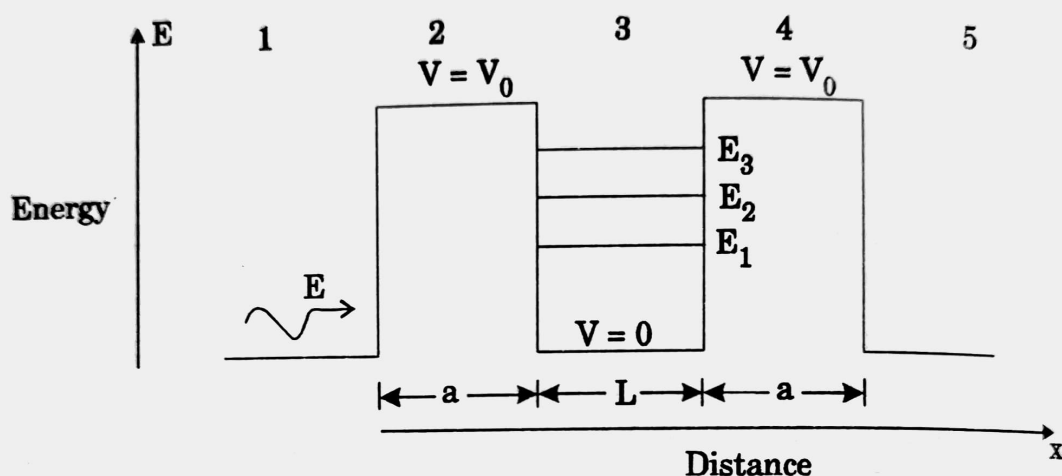


Fig. 5.8 Double symmetric barrier structure

The barriers are sufficiently thin to allow tunneling and the well region between the two barriers is also sufficiently narrow to form discrete (quasi-bound) energy levels, as shown in fig. 5.8.

The analysis of the double barrier structure is essentially the same as considered for single barrier tunneling.

The transmission coefficient of the double symmetric barrier becomes unity (ie., $T = 1$), when the energy of the incoming electron wave (E) coincides with the energy of one of the discrete states formed by the well.

$$\text{ie., } E = E_n = \frac{n^2 h^2}{8 m_e L^2}$$

where $n = 1, 2, 3 \dots$

Thus, transmission probability of the double symmetric barrier is maximum and hence, the tunneling current reaches peak value when energy of electron wave is equal to quantised energy state of the well.

This phenomenon is known as resonance tunneling.

The double barrier tunnel junction has important applications to a device known as a **resonant tunneling diode**. The working of these diodes can be understood considering the influence of bias on the energy band diagrams for the double barrier system.

When the incident electron energy E is very different from the energy of a discrete state E_n , transmission is low. As E tends to E_n , transmission will increase, becoming a maximum when $E = E_n$.

For example, assume that incident electrons have energy E , and that, all the energy states E_n lie above E , as shown in fig. 5.8.

As E increases, tunneling will increase, reaching a peak when $E = E_1$. After that point, a further increase in E will result in a decreasing current, as shown in fig. 5.9.

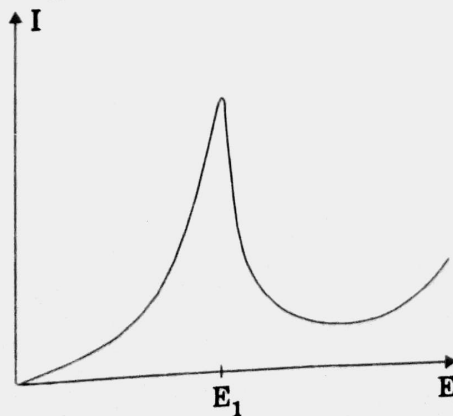


Fig. 5.9 Current - Energy characteristic for a resonant tunneling junction

This decrease of current with an increase of bias is called **negative resistance**. Further, peaks and valleys will occur as E approaches, and then moves **across**, other discrete energy states.

Resonant Tunneling Diode (RTD)

A typical resonant tunneling diode structure is made by using n -type GaAs for the regions to the left and right of both barriers (regions 1 and 5) in fig. 5.10 intrinsic GaAs for the well region (region 3) and AlGaAs or AlAs for the barrier material (regions 2 and 4 in fig. 5.10).

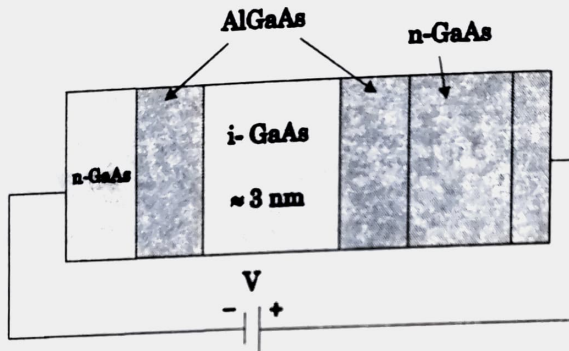


Fig. 5.10 Structure of Resonant Tunnel Diode (RTD)

Tunneling is controlled by applying a bias voltage across the device.

5.5 SINGLE ELECTRON PHENOMENA

In electronics, transistor is the most important device. Transistors are what computers use to compute-tiny switches turning ON and OFF making logic decisions.

Today, microchips have over a billion transistors, each one turning ON and OFF a billion times every second.

These chips require manufacturing processes with roughly sub 100-nanometre resolution. Every year, this technology resolution drops enabling even smaller transistors. Thus, more transistors are squeezed into the same amount of semiconductor space.

Interestingly, when each transistor is reducing to a few atoms, or a single molecule, quantum effects will play a significant role.

In 1970, to switch ON a silicon transistor, it required about 10 million electrons. Present day, transistors require closer to 10,000 electrons. Rather than moving many electrons through transistors, it may very well be practical and necessary to move electrons one at a time.

The single electron devices are sensitive to the transfer of even single electron charge. Single electron devices provide a potential application of ultra large scale integrated circuits with device size in the order of nanometres. They exhibit high speed operation with low power dissipation.

Coulomb – Blockade Effects

As the size of the quantum dot decreases, the charging energy W_c of a single excess charge on the dot increases.

If the quantum-dot size is sufficiently small and the charging energy W_c is much greater than thermal energy kT , no electron tunnels to and from the quantum dot.

Thus, the electron number in the dot takes a fixed value, say zero, when both the electrodes are grounded.

Definition

The charging effect which blocks the injection or rejection of a single charge into or from a quantum dot is called *Coulomb blockade effect*.

Condition for coulomb blockade

If two or more charges near one another, they exert coulomb forces upon each other. If two charges are the same kind, the force is repulsive. Therefore, the condition for observing coulomb blockade effects is expressed as

$$W_c = \frac{e^2}{2C} \gg kT$$

where C – capacitance of the quantum dot

T – temperature of the system.

W_C – charging energy and this is the energy needed to add one negatively charged electron to the dot.

From the above equation (1), we can notice that W_C is inversely proportional to the quantum dot's capacitance.

Thus, a larger capacitor can quite accommodate another electron without too much energy required. In the contrary, with extremely small capacitors, ($C \approx 10^{-19}$ F) like quantum dots, the charging energy is substantial and it can be large enough to “block” tunneling electrons as shown in fig. 5.11.

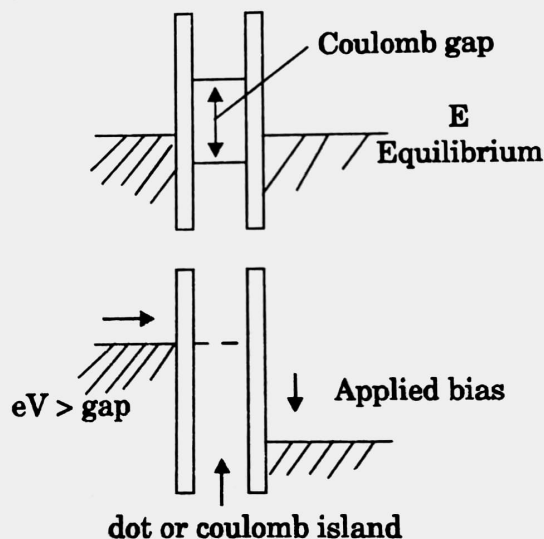


Fig. 5.11 Energy gap in Coulomb blockade

By referring fig. 5.12, it should be noted that by applying a positive bias to the gate electrode, an electron can be attracted to the quantum dot. The increase of the gate voltage attracts an electron more strongly to the quantum dot.

When the gate bias exceeds a certain value an electron enters quantum dot and the number of electron in the dot becomes one.

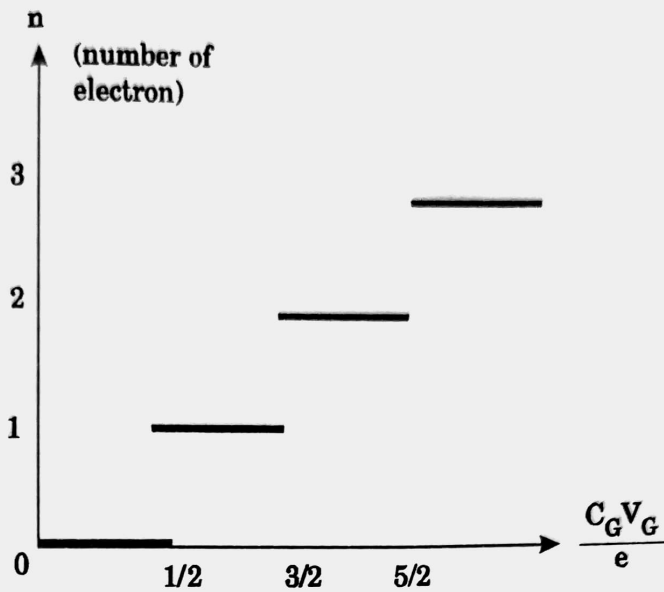


Fig.5.12 Electron number versus gate voltage characteristics of single-electron box.

Further, increase of the gate voltage makes it possible to make the electron number two and so on as in fig. 5.6. Thus, in the single-electron box, the electron number of the quantum dot is controlled, one by one, by utilizing the gate electrode.

Single Electron Tunneling

Tunneling is the way the electrons cross both the physical barriers and the energy barriers separating a quantum dot from the bulk material that surrounds it.

If any number of electrons on one side of the barrier could just tunnel across it, there would not be any isolation.

So it is necessary to control the addition and removal of electrons in a quantum dot.

When the size gets reduced, the capacitance also reduces to a small value. (Fig. 5.13)

At small sizes, the energy required to store an additional electron on it, $W = \frac{Q^2}{2C}$, may become larger than the thermal energy kT .

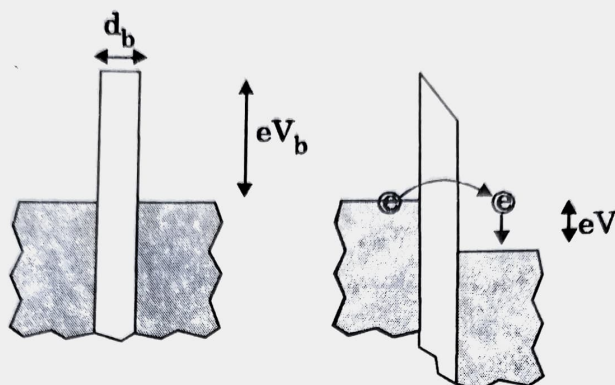


Fig. 5.13 Schematic representation of an electron tunneling through a barrier

As a consequence, the quantization of charge can dominate and tunneling of single electrons across leaky capacitors carries the current. This is called **single electron tunneling**.

It is used to design new types of devices, for example single-electron transistor using quantum dots.

So single-electron device are devices that can control the motion of even a single electron and consist of quantum dots which have tunnel junctions.

5.6 SINGLE ELECTRON TRANSISTOR (SET)

Although a single-electron box can control the number of electrons in the quantum dot, it does not have the properties of a switching device.

Single switching devices are essential elements in **Ultra large scale integrated circuits (ULSIs)**. The single-electron switching devices are required to utilize single electron devices in logic circuits.

Definition

SET is three-terminal switching device which can transfer electrons from source to drain one by one.

That is the purpose of the SET is to individually control the tunneling of electrons into and out of the quantum dot.

Construction & working

The structure of SET is shown in fig. 5.14. It has the similar structure of a conventional Field Effect Transistor (FET).

It has tunneling junctions in place of pn-junctions and quantum dot in place of the channel region of the FET.

To control tunneling, a voltage bias to the gate electrode is applied. A separate voltage bias is applied between source and drain electrodes for the current direction.

For current to flow, gate bias voltage must be large enough to overcome the Coulomb blockade energy.

The energy E needed to move a charge Q , across a potential difference V is given by

$$E = VQ.$$

Presently, the charge of an electron ($Q = e$). So, the voltage that will move an electron onto or off the quantum dot is given by

$$\therefore V = \frac{E}{e} = \frac{W_c}{e}$$

$$\text{Since } E = W_c,$$

where W_c – charging energy

$$V = \frac{\frac{e^2}{2C}}{e} = \frac{e}{2C} \quad \dots(1) \quad \left[\therefore W_c = \frac{e^2}{2C} \right]$$

With this voltage applied to the quantum dot, an electron can tunnel through Coulomb blockade of the quantum dot. The gate voltage as given in eqn.(1) can be used to tune the number of electrons on the dot at a time.

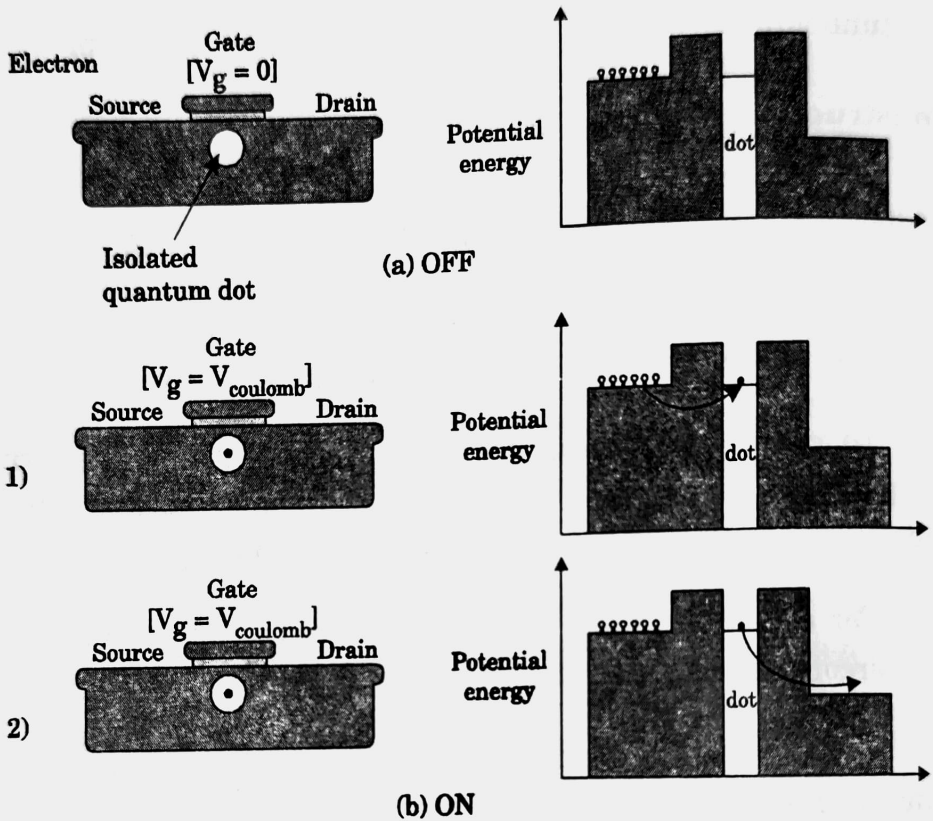


Fig. 5.14. A single-electron transistor (SET)

(a) SET in "OFF" mode. (b) SET in "ON" mode.

At the lowest setting, electrons tunnel one at a time, from source to drain through the dot. At the proper gate voltage, $V = \frac{e}{2C}$ the potential energy of the dot is low enough to allow an electron to tunnel through Coulomb blockade energy barrier to the quantum dot.

Once the quantum dot gets an electron, its potential energy rises. Then the electron tunnels through the Coulomb blockade on the other side, reach the lower potential energy at the drain.

Since, the dot is empty and potential is lower again, the process repeats.

Thus, when the gate voltage V_g is zero, no current flows. The first gate voltage which is large enough to move an electron through the Coulomb blockade is called V_{Coulomb} .

For single-electron tunneling, $V_g = V_{\text{Coulomb}}$.

If the gate voltage equals $V_{\text{Coulomb}} + \frac{e}{2C}$, then two electrons can be moved on the quantum dot at a time. If the gate voltage equals

$$V_{\text{Coulomb}} + \frac{e}{2C} + \frac{e}{2C} = V_{\text{Coulomb}} + \frac{e}{C},$$

then three electrons can be moved on the quantum dot at a time and so on.

Thus, number of electrons in the quantum dot is controlled using the gate voltage.

These ON and OFF states can be utilized to make an effective switch out of a SET.

The gate voltages for a SET are a few millivolts and source-to-drain currents are in the pico ampere range.

Advantages

- The fast information transfer speed between cells (almost near light speed) is carried out via electrostatic interactions only.
- No wire is needed between arrays. The size of each cell can be as small as 2.5 nm. This made them very suitable for high density memory.
- This can be used for the next generation quantum computer.

Limitations

- In order to operate SET circuit at room temperature, the size of the quantum dot should be smaller than 10 nm.
- It is very hard to fabricate by traditional optical lithography and semiconductor process.
- The methods must be developed for connecting the individual structures into logic circuits and these circuits must be arranged into larger 2D patterns.

Applications

- The main fields of application of the single electron transistor is used in sensor technology and digital electronic circuits.
- A variety of digital logic functions, including AND or NOR gates, is obtained based on SET operating at room temperature.
- It is used for mass data storage.
- It is used in highly sensitive electrometer.
- SET can be used as a temperature probe, particularly in the range of very low temperatures.
- SET is a suitable measurement set-up for single electron spectroscopy.
- It is used for the fabrication of a homo-dyn receiver operating at frequencies between 10 and 300 MHz.

5.7

MESOSCOPIC STRUCTURE

Mesoscopic structure deals with materials of an intermediate length. The scale of these materials can be described as being between the size of a quantity of atoms (such as a molecule) and of material containing micrometers.